Commodity	2016		2017	
	Price	Quantity	Price	Quantity
A	2	8	4	6
B	5	10	6	5
C	4	14	5	10
D	2	19	2	13

Solution:

Calculation of various indices

Commodity	2016		2017		p_1q_0	p_0q_0	p_1q_1	p_0q_1
	Price p_0	Quantity q_0	Price p_1	Quantity q_1				
A	2	8	4	6	32	16	24	12
B	5	10	6	5	60	50	30	25
C	4	14	5	10	70	56	50	40
D	2	19	2	13	38	38	26	26
					$\sum p_1q_0 = 200$	$\sum p_0q_0 = 160$	$\sum p_1q_1 = 130$	$\sum p_0q_1 = 103$

(1) Laspeyre's Index:

$$P_{01}^{L} = \frac{\sum p_1q_0}{\sum p_0q_0} \times 100$$

$$= \frac{200}{160} \times 100 = 125$$

(2) Paasche's Index

$$P_{01}^{P} = \frac{\sum p_1q_1}{\sum p_0q_1} \times 100$$

$$= \frac{130}{103} \times 100 = 126.21$$

(3) Dorbish and Bowley's Index

$$P_{01}^{DB} = \frac{P_{01}^{L} + P_{01}^{P}}{2} = \frac{125 + 126.21}{2}$$

$$= 125.6$$

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